



Localizing a Centerline File for MARC Rate Development

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

**MARC
Management**

1.0	Introduction	1
2.0	Best Practice Description.....	1
3.0	Implementation Steps	2
4.0	Benefits.....	4
5.0	Resources Required	4
6.0	Supporting Attachments / References	4
7.0	Related Best Practices	5
8.0	Acknowledgements.....	5

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1.0 Introduction

The Global (sometimes referred to as factory) LCC (Life Cycle Cost) maintenance and repair centerline is a theoretical baseline of operating cost parameters, derived collaboratively with each product group, and based on medium application. The sole safe-source for these files include the CGM Builder downloads page (for Builder Centerline files), and the AMT production server for AMT Centerline files). Most expanded mining product centerlines are built using AMT software, and traditional CGM machines are built using Builder software. Regional MARC Consultants (RMC's) will have access to the latest AMT Centerline files, and Builder files are available for download.

This LCC centerline development process is a critical input to support any O&O (Owning and Operating), or TCO (Total Cost of Ownership) cost estimation. Operating cost is comprised of energy requirements, repair cost (including undercarriage), planned maintenance cost, and special wear items. Reference the latest edition of the O&O chapter found in the Caterpillar Inc. performance handbook for a comprehensive list of criteria. LCC Centerline development provides baseline maintenance and repair cost per hour rate calculation in support of CSA (Customer Support Agreements) and MARCs (Maintenance And Repair Contracts).

An LCC centerline is a collection of tasks that define equipment servicing activities (minor/major repairs, overhauls, inspections, maintenance, replacement, etc.). These strategy tasks or repair options are performed at pre-determined design life or "time" intervals (measured in machine hours, tons, calendar, etc.) on a given equipment, assembly, or a component. Costs of the tasks are derived from service task job operations, detailing the scope, parts, labor, and any miscellaneous requirements associated with the servicing task.

Global LCC centerlines should **not** be confused with or used verbatim in site, customer or application specific needs, nor should it be construed as a MARC (Maintenance and Repair Contract) rate. These centerlines should be used as a starting point during the localization process.

2.0 Best Practice Description

The objective of this MARC rate localization process is to build rates that are:

- Machine, customer and mine-site specific.
- Realistic and achievable.
- Reflective of what a dealer can deliver.

This document provides process guidelines to support localization (modification) of global LCC centerline files. This will ensure that the rate accurately reflects the repair strategy of the Dealership, the configuration of the machine, and that the Dealer can deliver what is specified in the centerline file. This process will result in reasonable estimates in support of quoting a MARC contract (no attempt is made or implied to provide precise hourly costs). This best practice will also include examples of the mechanisms of how to accomplish localization using Builder G3 and AMT software.

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Localizing a Centerline File for MARC Rate Development	DATE 2/20/2015	CHG NO 02	NUMBER 0907-3.3-1086

The centerline user community should use the global LCC centerline file as a starting point, but it will need to be localized/modified to achieve a realistic, regional, site specific and dealer attainable LCC rate to suit equipment and site conditions.

3.0 Implementation Steps

Under **no** circumstance should a global LCC centerline file be used as a MARC rate without reviewing each strategy task / repair option and by selecting the appropriate strategy task / repair option for the application.

Ensure users start with the most current LCC centerline available. Changes/updates to the global LCC centerlines will be communicated to the Dealers, regions and to RMCs and the centerline files will be posted either on mining.cat.com builder downloads page or on the AMT production server (contact your local RMC for access to the latest AMT Centerline files). The dealer connect site “Maintenance and Repair Contract (MARC) Management” will be the primary method used to facilitate communications of updates, exchange information, and transfer LCC centerline change knowledge among LCC centerline users and stakeholders.

It is imperative that all stakeholders join this Dealer connect site in order to receive the information. This process will ensure global centerline change knowledge is distributed appropriately to regions, RMC’s and to Dealers.

In addition to using the global LCC centerline as a start, there are several localization aspects that must be reviewed. The following localization criteria must be reviewed for each repair option / strategy task and modified as necessary to ensure the Dealership can deliver within the CSA /MARC proposal:

- **Selecting the appropriate repair options/strategy tasks**

Certain major and minor assemblies will have different repair options available. For instance, a 797F mining truck’s engine may have the option of recondition or replacing with Reman. Both repair options/strategy tasks will be listed on the Centerline. In order to choose the appropriate strategy task for the application, the user must know the preferred repair strategy of the Dealership for major and minor components. Note: Some LCC files contain alternate and redundant repair options that could inflate the MARC rate if a single preferred repair strategy is not selected. This will inflate costs if alternate and redundant options are included (in addition to the preferred repair option). For example, two mutually exclusive strategies may exist at the same service interval and either the ‘Recondition before Failure’ or the ‘Replace with Cat Reman’ strategy task must be used – not both since they are conflicting.

In order to choose the appropriate Repair Options/strategy tasks for the application, the user must know the actual machine configuration in terms of mandatory and optional attachments. Optional attachments or optional configurations are identified by “If Equipped (IEQ)” in the job condition field in Builder G3 or by cost bearer selection in AMT. “External” cost bearer will indicate that the strategy task is an option / or typically excluded from contract rates. If it should be included in any rate calculation, it must be changed to “Contract”. A Dealer may use the actual Machine Shipping Order (MSO), or may retrieve the machine consist from SIMSi (Service Information Management System) at the following web address:

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Localizing a Centerline File for MARC Rate Development	DATE 2/20/2015	CHG NO 02	NUMBER 0907-3.3-1086

<https://sims.cat.com/sims/simsi/index.jsp> A PDF example of what the SIMSi Product Configuration search will produce is provided in the 'Supporting Attachments/References' section of this Best Practice.

- **First / Frequency Interval hours - PCR (Planned Component Replacement)**

It is imperative to use realistic and achievable component lives. Artificial PCR (Planned Component Replacement) life inflation (without sufficient data to corroborate) which lowers the rate should NEVER be used to gain financial competitiveness. An acceptable rate reduction strategy should consider tenants of Rate Commercialization which is an acceptable CGM mechanism in place to support competitive bids. Poor decision strategies such as artificially inflating PCR lives may lead to unprofitable business decisions, and ultimately brand erosion since component lives will most likely not achieve expectations.

While not all PCR deviations need to be reviewed by the global MARC teams, guidance for PCR life increases which requires a global MARC team review may include (but is not limited to) any change that will reduce the scope by at least one rebuild/overhaul during the term of the contract.

- **Labor Hours**

Due to regional dealer variances and varying skill sets, the global centerline labor hours will need to be reviewed and may need to be modified to suit local competency levels, and number of tradespeople per repair option. The centerline labor hours (Duration and Labor Hours) for R&I and Recondition jobs are estimates and should be reviewed and adjusted based on local expertise and experience.

- **Labor Rates**

Ensure the labor rates in the Builder Labor Table / AMT quick editor labor rates are configured for your Dealership's rates. You should review the total costs associated with each Repair Option / Strategy Task to make sure your Dealership can deliver at this cost. It is each Dealer's responsibility to configure the rates with established Field/Shop labor rates.

- **Parts replacement percentages**

Based on local conditions, application, repair strategy, maintenance and operating practices, individual part replacement percentages may need to be adjusted from the global centerline position. This is based on local expertise and experience.

- **Dealer part pricing/part identification**

Obtain the appropriate regional dealer pricing file(s), and Import the latest parts pricing into your centerline file. This provides DN (Component Unit Cost) and SCL (Component Unit List) for that specific region. CAT Global Mining Parts & Pricing (CGMPP), and the local Dealer price tapes are the sole safe-sources for CGM parts pricing information. Next Gen SIS and SIS web are sole safe-sources for CGM parts data (excluding pricing), drawings and identification.

- **Salvage opportunities**

CGM Centerline files developed with Caterpillar engines include Job Operations within the Engine Overhaul Repair Options for salvage work for the cylinder block and crankshaft. Engine

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Localizing a Centerline File for MARC Rate Development	DATE 2/20/2015	CHG NO 02	NUMBER 0907-3.3-1086
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block and engine crankshaft Salvage costs are provided in the centerline files based on actual dealer cost data. A spreadsheet (found on the Builder downloads page in the supporting documents section) details actual DN costs of the crank and block for specified engines. The source CRC data was used to provide accurate cost for salvaging the engine block and crankshaft for both first and second overhaul. Use the existing actual cost structure if you have no machine history on salvage. You will need to modify the % replacement of the components to reach the average actual CRC costs identified in the spreadsheet. If your dealership has accumulated statistically significant sample size, change the % replacement value to match your current actual salvage cost.

Not all Builder files have been updated to reflect the salvage costs. Use the spreadsheet on the Builder downloads to represent the salvage costs to the engine block and crankshaft.

- **Custom Repair Options**

Add any custom Repair Options / Strategy Tasks that may be required. For example, if there are certain regulatory / safety requirements needed in your region, a 'Safety Inspection' might need to be included and performed every 250 hours and assigned 2 hours of labor to the job since it is a required local regulatory requirement. Alternately, if it is customary for your Dealership to perform CM (Condition Monitoring) services as part of the CSA arrangement, these CM repair options / strategy tasks should be added to the centerline.

The attached document illustrates various examples of how to tailor the centerline Builder or AMT file in order to:

- Match the machine configuration sold to the customer.
- Reflect the repair strategy of the Dealership for major and minor components.
- Review and edit labor hours, PCR intervals, parts replacement percentages, salvage opportunities, etc.

To implement this Best Practice, apply the same logic that is outlined in the examples within the supporting PowerPoint documents.

4.0 Benefits

The Benefit of applying this process when localizing global LCC centerline files is assurance that your MARC or other CSA rate will be:

- Machine and mine-site specific.
- Realistic and achievable.
- Reflective of what a dealer can deliver.

5.0 Resources Required

AMT and or BuilderG3 software.

6.0 Supporting Attachments / References

- *Estimating Owning and Operating Costs.pdf* - Caterpillar Performance handbook,
- *CGM Centerline File Localization.pptx* – Instructions contained in this Best Practice in PowerPoint format
- *SIMS Product Configuration Example.pdf* – Sample product consist printout from SIMSi.

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Localizing a Centerline File for MARC Rate Development	DATE 2/20/2015	CHG NO 02	NUMBER 0907-3.3-1086

7.0 Related Best Practices

0906-5.05-1009 MARC Rate Development
1006-3.3-1024 Ensuring MARC Profitability
0807-3.3-1085 Updating builder files with current parts

8.0 Acknowledgements

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